LED-Based Water Level Detection System

Project Report · Digital Logic Design

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1. Introduction

Water level monitoring is something we need in a lot of everyday situations, whether it is a household water tank, an overhead storage tank, or even an industrial reservoir. If the water level is not tracked properly, it can overflow and cause damage, or drop too low and disrupt whatever depends on it. I wanted to build something simple that could visually show the water level at a glance using LEDs.

For this project, I designed and built a basic LED-based water level detection system. The core idea was to use three BC547 NPN transistors as switches, where each transistor controls one LED. When water touches a probe placed at a certain height inside the container, it completes the base circuit of the transistor and turns the corresponding LED on. I used three probes to represent three levels: low, medium, and high.

I assembled everything on a breadboard and powered the circuit with a 9 V battery. It was a straightforward build, but it taught me a lot about how transistors actually behave in a real circuit, how to pick the right resistor value to protect LEDs, and how small signals at the base of a transistor can control a larger current through the collector. Overall, it connected a lot of the theory from Digital Logic Design to something I could physically see working.

2. Objectives

- Build a functional water level detection circuit using transistors as electronic switches.
- Understand and demonstrate how a BC547 NPN transistor works in switching mode.
- Use LEDs of different colours to indicate three water level states: Low, Medium, and High.
- Use current-limiting resistors to keep the LEDs from burning out.
- Get hands-on practice with breadboard wiring and basic DC circuit calculations.

3. Components and Materials

Below is a list of all the components I used to build this circuit:

Component	Quantity	Purpose / Function
Breadboard	1	Used to assemble and test the circuit without soldering
9 V Battery with Holder	1	Powers the entire circuit
BC547 NPN Transistor	3	Acts as an electronic switch to turn each LED on or off
LEDs (Red, Yellow, Green)	3	Indicate low, medium, and high water levels respectively
330 Ω Resistors	3	Limit the current flowing through each LED
Jumper Wires	As needed	Connect all the components together on the breadboard

4. Working Principle

4.1 Transistor as a Switch

The BC547 is a general-purpose NPN transistor. In this circuit, I connected each transistor in common-emitter configuration. When the water level rises enough to touch a probe, it creates a small base current that drives the transistor into saturation. Once that happens, a much larger current flows from the collector to the emitter, which is enough to light up the LED connected to that stage. Essentially, the water itself acts as the trigger.

4.2 Current-Limiting Resistors

I placed a 330 Ω resistor in series with each LED between the collector and the +9 V supply. Without this resistor, too much current would flow and burn the LED out. Using a quick Ohm's Law calculation: $I = (9\ V - 1.8\ V) / 330\ \Omega \approx 21.8\ mA$, which is safely within the LED's rated range. So the resistor both protects the LED and keeps the current at a level where the brightness looks right.

4.3 How the Level Detection Works

I placed three conductive probes inside the water container at different heights, and each one connects to the base of a different transistor. Here is what happens at each level:

- Low Level Probe — turns on the Red LED when water first reaches the bottom probe.
- Medium Level Probe — turns on the Yellow LED when water reaches the middle probe.
- High Level Probe — turns on the Green LED when water reaches the top probe.

Since water conducts electricity, it completes the base circuit of the transistor the moment it touches the probe. The transistor then switches on and lights the LED. When the water drops back below a probe, the base current disappears, the transistor goes back to cut-off, and the LED turns off.

5. Circuit Diagram

The diagram below shows how the three transistor stages are connected. All three BC547 transistors (Q1, Q2, Q3) share a common ground. The collector of each one connects to a 330 Ω resistor and then to an LED, with the other end of the LED going to the +9 V rail. The probes are wired to the base of each transistor and dipped into the water at the three height marks.

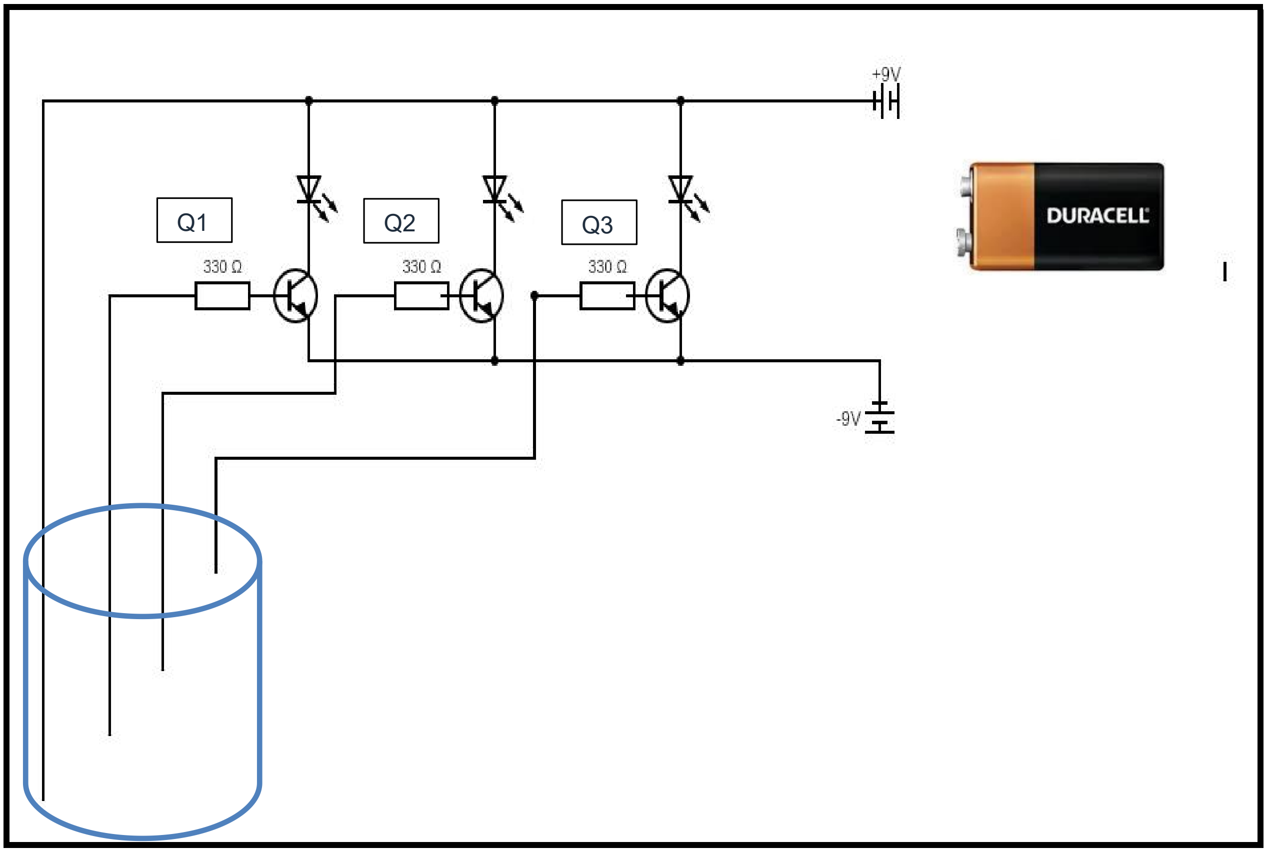

Figure 1: Circuit schematic of the LED-based water level detection system

6. Results and Observations

Once I assembled the circuit and tested it, it worked exactly as I expected. I tested four different conditions by gradually adding water to the container and noting which LEDs came on. Here are the results I observed:

Test Condition	Active Transistor(s)	LEDs Illuminated	What I Observed
No water / dry probes	None	None	All LEDs stayed off as expected
Water at low probe only	Q1	Red	Red LED turned on; low level detected correctly
Water at low and mid probes	Q1, Q2	Red, Yellow	Both LEDs on; medium level confirmed
Water touching all three probes	Q1, Q2, Q3	Red, Yellow, Green	All three LEDs on; high level indicated correctly

No component failed during testing. The resistors did their job well, and the LEDs lit up at a good brightness without getting too hot. The circuit also responded right away whenever the water level changed, which showed that the transistors were switching properly.

7. Discussion

Overall, I think the circuit worked really well for what it was designed to do. One thing I noticed is that the design can easily be expanded. If someone wanted more resolution, they could just add more probes, transistors, and LEDs at additional heights. The stages are independent of each other, so adding more does not break what is already there.

One thing I thought about after building it is that water conductivity is not always the same. Tap water works fine because it has dissolved minerals that help it conduct, but very pure or distilled water might not give enough base current to fully saturate the transistor. In that case, a small op-amp comparator could be added before the transistor base to boost the signal. That would make the circuit more reliable in different water conditions.

Another possible improvement would be adding a buzzer or a relay that activates when the water reaches the high level, so the system could also produce an alert or automatically cut off a pump. Right now it is purely visual, but with a small addition it could become a more complete control system. These are the kinds of extensions I would want to explore if I revisited this project.

8. Conclusion

This project was a really useful exercise for me. I built a three-stage water level detection circuit using BC547 transistors and LEDs, and it worked correctly across all the tests I ran. Each probe triggered its corresponding LED at the right time, and no components were damaged throughout the process.

More than just making something work, this project helped me understand transistors in a way that reading about them in a textbook does not fully capture. I could actually see how a tiny bit of base current switches on a much larger current, and how that translates to a visible output. It also reinforced how important it is to calculate resistor values properly before connecting anything. I feel a lot more confident working with circuits after completing this.

References

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